

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

Nov/Dec 2012

EE303 Microcontrollers and their Applications

Date: 28.11.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Draw block diagram of 8051 and explain it in detail. [05]
(b) Draw pin diagram of 8255 [02]
- Q - 2 (a) List out all possible difference between 8051 and 8085 [10]
(b) Generate Fibonacci series up to N terms and save from location 50h onwards. Assume N being available in location 4Fh. Also assume the value of N to be greater than 3. [04]

OR

- Q - 2 (a) Explain in detail ADD instruction [05]
(b) 20 bytes of data are stored in location 7Fh to 6Ch of internal RAM. Count the number of those bytes, which contain 00h, and store this number of null bytes in RAM location 6Bh. [05]
(c) Find out the number of 1's in a byte, available in internal data memory location 50h. Store the result in the accumulator itself. [04]
- Q - 3 (a) Convert the following hex numbers to binary and decimal [04]
a) B96H b) C43H c) 7EH d) FA25H
(b) Do as directed: [05]
1) Draw the logic diagram for $Y = AB + CD$
2) The 8052 has _____ bytes of on-chip ROM.
3) The 8051 family has _____ pins for I/O.
4) What is the major difference between 8051 and 8052 microcontrollers?
5) The 8051 is an _____ bit microprocessor
(c) There are 25 prime numbers between 2 and 100. Find a method to generate these prime numbers. [05]

OR

- Q - 3 (a) Explain in detail about TCON a special function register [05]
(b) Assuming crystal frequency to be 22MHz, write a program to generate a square wave of 1KHz on P1.7 [05]
(c) Subtract the contents of RAM location 13h from RAM location 2Bh put the result in RAM location 3Ch. [04]

SECTION - II

- Q - 4 (a) Write a program to write "HELLO THERE" on 2x16 LCD [07]
Q - 5 (a) Write a program to generate a sine wave [05]

- (b) Write a program to generate two square waves – one of 5KHz frequency at pin P1.3 and another of frequency 6KHz at pin P2.3. Assume XTAL = 22MHz (use interrupt method to create program) [05]
- (c) Two packed BCD digits are available in location 30h (MSB) and 31h (LSB). Write a program to pack these two and store in register R7. [04]

OR

Q - 5 (a) Do as directed: [14]

- i. Find the contents of register A after execution of the following code
CLR A
ORL A, #99h
CPL A
- ii. What value must R4 have in order for the following instruction not to jump?
CJNE R4, #53, OVER
- iii. If CY=1, A=95h and B=4Fh prior to the execution of "SUBB A, B", what will be the contents of A after the subtraction?
- iv. Is this a valid instruction in 8051? "DIV A, R4". Justify
- v. To configure 8051 as input ports which value is to be sent on that port?
- vi. Show how to save status of P2.7 in RAM bit location 31.
- vii. List any two assembly directives.
- viii. Indicate selection made in the statement "MOV TMOD, #20h".
- ix. State the features of IC 8752.
- x. Find the machine cycle for a crystal frequency of 12MHz.
- xi. Which port in 8051 does not have in build pull up resistor?
- xii. In mode 1 counter rolls over when it goes from ____ to ____.
- xiii. What addresses in the IVT is assigned to INT0 and INT1?
- xiv. What is the value of A after each of the following code?
MOV A, #45H
CLR A
SETB C
RRC A
SETB C
RRC A

Q - 6 (a) Write a short note on instruction CJNE [04]

(b) Draw pin diagram of 8051 and state the features of architecture of 8051 family [06]

(c) Do as directed; [04]

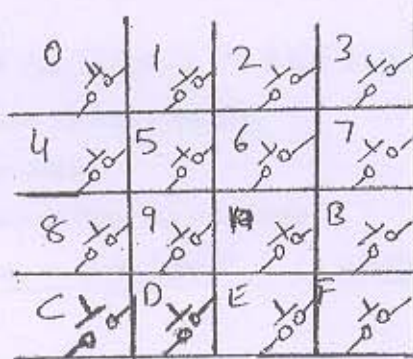
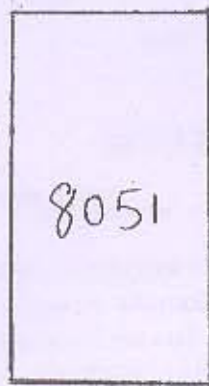
- i. Output of DAC is in ____ (current, voltage)
- ii. In ADC 0804, the INTR signal is an ____ (input, output)
- iii. To see if any key is pressed, all rows are grounded. (True or False)
- iv. RS pin in LCD is a ____ (input, output) pin.

OR

Q - 6 (a) An array of 10 bytes is available in the external data memory location from 2000h onwards. Compare it with the array of 10 bytes available in the internal data memory location from 30h onwards. If both are identical in all respects, then return with carry (CY) flag cleared; otherwise set CY flag at return. [05]

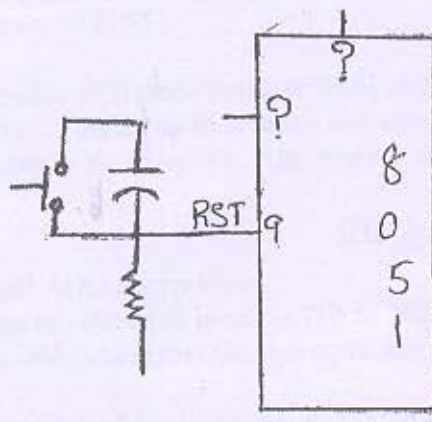
(b) Rectify the mistakes in circuit diagram. If in some cases the circuit is incomplete, complete it. [09]

①



4x4 Matrix Keyboard.

②



③

